

Assignment: Fundamentals of Linear Algebra for Artificial Intelligence

Instructions

- Answer all the questions.
 - Show all intermediate calculations wherever applicable.
 - Use a calculator if required.
 - Round numerical answers to **two decimal places** where necessary.
 - Clearly write the final answer for each question.
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Part A – Scalars, Vectors and Matrices

Question 1

Identify whether each of the following is a **Scalar**, **Vector**, **Matrix**, or **Tensor**.

a)

$$5$$

b)

$$\begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}$$

c)

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

d)

A colour image of size

$$256 \times 256 \times 3$$

Question 2

Determine the dimensions of the following matrices.

a)

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

b)

$$\begin{bmatrix} 5 \\ 8 \\ 9 \\ 2 \end{bmatrix}$$

c)

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$$

Question 3

Given

$$A = \begin{bmatrix} 2 & 5 \\ 3 & 7 \end{bmatrix}$$

and

$$B = \begin{bmatrix} 4 & 1 \\ 6 & 8 \end{bmatrix}$$

Find

- $A + B$
- $A - B$
- $2A$

Question 4

Find the transpose of the matrix

$$A = \begin{bmatrix} 2 & 4 & 6 \\ 8 & 10 & 12 \end{bmatrix}$$

Question 5

Find the transpose of

$$\begin{bmatrix} 1 & 3 \\ 2 & 5 \\ 4 & 6 \end{bmatrix}$$

Part B – Matrix Multiplication

Question 6

Multiply

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

and

$$B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

Question 7

Determine whether the following matrix multiplications are possible. If they are possible, write the dimensions of the resulting matrix.

a)

$$(2 \times 3)(3 \times 4)$$

b)

$$(3 \times 2)(4 \times 3)$$

c)

$$(4 \times 2)(2 \times 5)$$

d)

$$(5 \times 4)(4 \times 2)$$

Question 8

Given

$$A = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$$

and

$$B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

Find

$$AB$$

Question 9

Verify whether matrix multiplication is commutative.

Compute

$$AB$$

and

$$BA$$

where

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

and

$$B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

Based on your calculations, what do you conclude?

Part C – Vector Operations

Question 10

Find the magnitude of the vector

$$x = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

Question 11

Find the magnitude of

$$y = \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix}$$

Question 12

Compute the dot product of

$$x = \begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix}$$

and

$$y = \begin{bmatrix} 5 \\ 2 \\ 3 \end{bmatrix}$$

Question 13

Determine whether the following vectors are orthogonal.

$$u = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$v = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

Question 14

Normalize the vector

$$x = \begin{bmatrix} 6 \\ 8 \end{bmatrix}$$

Question 15

Find the Euclidean distance between

$$A = (2, 3)$$

and

$$B = (8, 11)$$

Part D – Vector Spaces and Span

Question 16

Determine whether the following vectors are linearly independent.

$$v_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$
$$v_2 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

Explain your answer.

Question 17

Determine whether

$$v = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$$

belongs to the span of

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

and

$$v_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Question 18

Find a basis for

$$\mathbb{R}^2$$

Question 19

Find the rank of the matrix

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

Question 20

Find the rank of

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Part E – Orthogonality

Question 21

State whether the following pairs of vectors are orthogonal.

a)

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

b)

$$\begin{bmatrix} 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 4 \\ -3 \end{bmatrix}$$

c)

$$\begin{bmatrix} 2 \\ 5 \end{bmatrix}, \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$

Question 22

Find the cosine similarity between

$$x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

and

$$y = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

Interpret the result.

Question 23

Explain, in your own words, the difference between:

- Orthogonal vectors
- Orthonormal vectors
- A basis
- A vector space

